

Dr. Dinesh Kumar

Curriculum Vitae

Mathematics & Basic Sciences
NIIT University, Rajasthan, India
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Employment

- June 2022 –Present **Assistant Professor**, *Mathematics & Basic Sciences*, NIIT University, Rajasthan, India
- Dec 2018 –June 2022 **Dr. D.S. Kothari Postdoctoral Fellow**, *Computational & Data Sciences*, Indian Institute of Science, Bengaluru, India
- Jan 2017 –Dec 2018 **NBHM Postdoctoral Fellow**, *Computational & Data Sciences*, Indian Institute of Science, Bengaluru, India
- Aug 2015 –Dec 2016 **Assistant Professor**, *School of Engineering & Technology*, BML Munjal University, India
- Jan 2015 –June 2015 **Assistant Professor (Contract)**, *Department of Mathematics*, National Institute of Technology Jalandhar, India
- Jul 2008 –Feb 2009 **Tutor**, *Department of Mathematics*, Seth Motilal (PG) College Jhunjhunu, India

Education

- 2015 **Doctor of Philosophy (Mathematics)**, *Indian Institute of Technology Guwahati*
- 2007 **Master of Science (Mathematics)**, *University of Rajasthan, Jaipur*
- 2005 **Bachelor of Science (Physics, Mathematics and Computer Application)**, *University of Rajasthan, Jaipur*

Ph.D. Thesis

Title ***Dynamics and Control of a Predator-Prey System with the Supply of Additional Food to Predators***

Abstract This thesis examines the efficient and important role of additional food in a predator-prey system. In this work, we derive and study a model for two species. The prey population is assumed to be growing logistically in absence of predators, with the predators having an additional food (non-prey) source apart from natural prey and the functional response is assumed to be ratio-dependent. This leads to the development of a modified ratio-dependent model. We study the modified ratio-dependent model via stability (local and global) analysis and analyze the consequences of providing additional food to the predators. By taking the spatial component into account, we consider a diffusive modified ratio-dependent model and obtain the necessary conditions for Turing instability to occur. We determine the role of additional food supply in the formation of Turing patterns. We also look into the effect of prey harvesting in the system with and without additional food supply to predators. This is accomplished by examining the stability analysis results and obtaining the optimal harvesting policy for both the ratio-dependent and modified ratio-dependent model with prey harvesting. Finally, we study a minimum time-optimal control problem for controlling pest(pre) population, where the quality of additional food is taken as the control variable.

In conclusion, thesis work provides the various significant applications of additional food supply in a predator-prey system, that includes population conservation, pest management (possible in the long term as well as minimum time), the formation of spatiotemporal patterns, and the determination of an optimal prey harvesting policy in conjunction with the population conservation, which is more efficient as compared to the case without an additional food supply.

Research Interests

My research is about understanding how biological or ecological systems work. I develop mathematical models of biological populations and analyze them using techniques from dynamical systems, bifurcation theory, spectral graph theory, and optimal control theory.

My postdoctoral work was about a novel problem of graph partitioning that deals with dynamic graphs. It has numerous applications; I focused on the (equilibrium) stability of dynamical complex networks and their partitioned subnetworks over which general reaction-diffusion systems run. By studying a mathematical model for metapopulations, I have investigated how and what factors contribute to the long-term stability of multi-species living and interacting in spatially-discrete patches. I established the connection between stability and connectivity of the metapopulations networks. I have developed some graph partitioning algorithms and obtained some theoretical results, which determine the spatial cuts that split the whole network into daughter subnetworks in which populations can survive.

Research Areas

Applied Mathematics (*Mathematical Modelling, Dynamical Systems & Bifurcation Theory, Spectral Graph Theory, Optimal Control Theory, and Computational Mathematics*)

• Mathematical Biology • Complex Networks • Graph Partitioning • Bioeconomic Optimization • Machine Learning Inspired Numerical Methods.

Research Publications

1. Kon, R., **Kumar, D.** (2023). Stability of Rosenzweig-MacArthur models with non-diffusive dispersal on non-regular networks. *Theoretical Population Biology*, **150**: 14–22. DOI: [10.1016/j.tpb.2023.02.002](https://doi.org/10.1016/j.tpb.2023.02.002) (IF: 1.3, Q2).
2. **Kumar, D.**, Abhishek, A., Raha, S. (2021). Stability aware spatial cut of metapopulations ecological networks. *Ecological Complexity*, **47**:100948. DOI: [10.1016/j.ecocom.2021.100948](https://doi.org/10.1016/j.ecocom.2021.100948) (IF: 3.4, Q2).
3. **Kumar, D.**, Gupta, J., Raha, S. (2019). Partitioning a reaction-diffusion ecological network for dynamic stability. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, **475**:20180524. DOI: [10.1098/rspa.2018.0524](https://doi.org/10.1098/rspa.2018.0524) (IF: 3.0, Q1).
4. **Kumar, D.**, Chakrabarty, S.P. (2018). A predator-prey model with additional food supply to predators: Dynamics and application. *Computational and Applied Mathematics*, **37**:763–784. DOI: [10.1007/s40314-016-0369-x](https://doi.org/10.1007/s40314-016-0369-x) (IF: 2.5, Q2).
5. **Kumar, D.**, Chakrabarty, S.P. (2017). Additional food induced Turing patterns for a diffusive predator-prey model. *International Journal of Applied and Computational Mathematics*, **3**: 165–183. DOI: [10.1007/s40819-015-0097-8](https://doi.org/10.1007/s40819-015-0097-8) (IF: 2.2, Q3).
6. **Kumar, D.**, Chakrabarty, S.P. (2015). A comparative study of bioeconomic ratio-dependent predator-prey model with and without additional food to predators. *Non-linear Dynamics*, **80**(1-2): 23–38. DOI: [10.1007/s11071-014-1848-5](https://doi.org/10.1007/s11071-014-1848-5) (IF: 5.6, Q1).
7. **Kumar, D.**, Chakrabarty, S.P. (2015). Optimal control of pest by supply of additional food to predators in a predator-prey system. *Journal of Advanced Research in Dynamical and Control Systems*, **7**(1): 56–65.

Preprints / Submitted Manuscripts

1. **Kumar, D.** (2025). On the Stability of Discrete Reaction-Diffusion System of Networked Dynamical Systems. *arXiv:2511.18798 [math]*. arxiv.org/pdf/2511.18798
2. **Kumar, D.** (2026). Sustainable and Optimal Harvesting in a Seasonally Harvested Fishery with a Marine Protected Area: A Two-Patch Model with Bang-Bang and Singular Control. *arXiv:2601.20563 [math]*. arxiv.org/pdf/2601.20563
3. **Kumar, D.** (Communicated). Clarifying Complex Differentiability: An Exposition.

Theses Supervised

- 2020 **Stability Preserving Bisection Algorithms in Reaction-Diffusion Complex Networks**, Vikram Bhatt, *M.Tech (Research)*, Indian Institute of Science, Bengaluru, India
- 2018 **Sustainable Partitioning Algorithm for Metapopulations Dynamics Over Network**, Jatin Gupta, *M.Tech (Course work)*, Indian Institute of Science, Bengaluru, India

Invited Talks & Seminars

- Nov. 29, 2021 **Stability Invariant Partitioning of Reaction-Diffusion Metapopulations Networks**, IIT Palakkad, India
- Aug. 24, 2021 **Stability Aware Cut of Metapopulations Complex Networks**, Muthayammal College of Arts & Science, Rasipuram (Tamil Nadu), India
- Dec. 26, 2020 **Stabilized Partitioning of Reaction-Diffusion Metapopulations Complex Networks**, IIT Roorkee, India
- Aug. 23–26, 2017 **Additional Food and Predator-Prey Dynamics**, 6th China India Japan Korea Mathematical Biology Colloquium Cum Conference, IIT Kanpur, India
- Oct. 30, 2012 **On Periodic Orbits**, Research Scholar Seminar Series, Department of Mathematics, IIT Guwahati
- May 31, 2011 **Local Bifurcations of Continuous Dynamical Systems**, Research Scholar Seminar Series, Department of Mathematics, IIT Guwahati

Workshops and Conferences Attended

- Nov. 21–30, 2012 **Advanced Workshop on Mathematical Theory of Control and Numerics**, Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram, India
- Feb. 25–Mar. 02, 2011 **Workshop on Stability & Bifurcation Analysis and Pattern Formation in Mathematical Ecology and Epidemiology**, Indian Institute of Technology (IIT), Kanpur, India
- Dec. 7–14, 2010 **Workshop and Symposium on Mathematical Ecology**, Indian Institute of Science Education and Research (IISER), Kolkata, India
- Aug. 30–Sep. 02, 2010 **International Workshop on Recent Advances in Computational Fluid Dynamics**, Indian Institute of Technology (IIT), Guwahati, India
- Jan. 18–19, 2009 **Rajasthan Ganita Parishad, 20th Annual Conference on Emerging Fields in Mathematics and their Interdisciplinary Uses**, Government College Kota, Rajasthan, India

Teaching Assignment at NIIT University

2023–24 & 2025–26 **MAT 101 (First year undergraduate B.Tech course on Linear Algebra and Differential Equations), NIIT University, India**

Differential Equations: First-order differential equations: separable equations, equations reducible to separable form, exact equations, and linear equations. Ordinary linear differential equations of second and higher order with constant coefficients: homogeneous and non-homogeneous equations, complementary function and particular integral, method of undetermined coefficients, method of variation of parameters, and Euler–Cauchy (equidimensional) equations. Laplace transforms: Laplace and inverse Laplace transforms, transforms of derivatives and integrals, first and second shifting theorems, unit step function, convolution of functions and convolution theorem, Laplace transform of periodic functions, and applications.

Linear Algebra: Fields, vector spaces and subspaces; linear dependence and linear independence; span, basis, and finite-dimensional vector spaces. Linear transformations; matrix representation of linear transformations; range and null space. Algebra of matrices; special types of matrices; rank of a matrix; elementary row operations; inverse of a matrix. Determinants: definition, properties, row and column operations, and evaluation of determinants. Solution of systems of linear equations using matrix methods. Eigenvalues and eigenvectors; Cayley–Hamilton theorem; diagonalization; applications of eigenvalue problems with emphasis on systems of differential equations. Inner product spaces; norms and their properties; orthogonality; Gram–Schmidt orthogonalization process.

Note: Each student team undertakes a mandatory industry-based project integrating differential equations and data structures, designed in collaboration with the parallel Data Structures course (CS course).

2023–24 & 2025–26 **MAT 112 (First year undergraduate B.Tech course on Calculus), NIIT University, India**

Infinite Sequences and Series: Limits of sequences and series, convergence of series, tests of convergence, power series, absolute and conditional convergence, radius of convergence, Taylor and Maclaurin series.

Partial Derivatives: Functions of several variables, limits, continuity, partial derivatives, chain rule, directional derivative, gradient vector, Taylor’s formula for two variables, extreme values and saddle points, Lagrange’s multiplier method.

Multiple Integrals: Double integrals in Cartesian and polar coordinates with applications (area, moments, etc.); triple integrals in Cartesian, cylindrical and spherical coordinates with applications (volumes, moments, etc.); change of order of integration and substitution in multiple integrals.

Vector Differential and Integral Calculus: Vector functions and their derivatives, arc length, tangent and normal vectors, gradient, divergence and curl, vector fields, line integrals, work, circulation, flux, independence of path, conservative fields, Green’s theorem, surface area, surface integrals, divergence theorem, Stokes’ theorem.

2022–25 **CS4101 / DS222 (Third/Second year undergraduate open elective: Introduction to Linear and Non-Linear Optimization / Optimization for Data Science), NIIT University, India**

Convex sets and functions, critical points, contour plots, gradient and Hessian of a function. *Unconstrained Optimization*: Simplex method, Fibonacci and golden section search, quadratic fit search, descent direction iteration, line search, approximate line search, gradient descent, conjugate gradient descent, stochastic gradient descent, Newton's method, subgradient method, coordinate descent. *Constrained Optimization*: Equality constraints, duality, Newton's method for equality constraints, interior point methods, barrier methods, penalty methods, cutting plane methods. *Metaheuristic Methods*: Genetic algorithms, differential evolution, particle swarm optimization, ant colony optimization, simulated annealing. *Python Lab (2 hrs/week)*: Computational implementation of all optimization methods above using Python.

2022–23 & 2024–25 **DS401 (Third year undergraduate Professional elective: Numerical Methods for Data Science)**, *NIIT University, India*

Fundamentals of Linear Algebra: Column space, multiplying and factoring matrices, orthonormal columns, eigenvalues, eigenvectors, positive definite and semidefinite matrices, singular values, singular vectors, principal components, best low-rank matrix, Rayleigh quotients, generalized eigenvalues, norms of vectors, functions and matrices.

Computation using Large Matrices: Numerical linear algebra, least squares, bases for the column space, survey of difficulties with $Ax = b$, minimizing $\|x\|$ subject to $Ax = b$, computing eigenvalues and singular values, randomized matrix multiplication.

Low Rank and Compressed Sensing: Matrices $A(t)$ depending on t , derivative $= dA/dt$, derivatives of inverse and singular values, rapidly decreasing singular values, counting parameters in SVD, LU, QR, saddle points.

Special Matrices: Completing a rank-one matrix, circulants, eigenvectors of circulant matrices, Fourier matrix, ImageNet, convolutional neural network (CNN), the convolution rule, neural nets and the learning function, distance matrices, Procrustes problem, finding clusters in graphs.

Statistics: Mean, variance, probability distributions, moments, cumulants, Markov chains.

Optimization: Maxmin principle, minimizing a function step by step, gradient descent: downhill to a minimum, accelerating gradient descent, linear programming and two-person games, stochastic gradient descent.

Learning from Data: Structure of neural nets for deep learning, backpropagation, finding partial derivatives.

2023–24 **MAT221 (Summer course on Probability and Random Processes for backlogs)**, *NIIT University, India*

2023–25 **MAT 011/012 (First year undergraduate course on Remedial Mathematics I & II for Biotechnology)**, *NIIT University, India*

2022–23 **MAT121 (First year undergraduate IMBA & BBA course on Business Statistics)**, *NIIT University, India*

Past Teaching Experience

2016 **MATH211 (Second year undergraduate course on Probability & Statistics)**, *BML Munjal University, India*

2016 **MATH112 (First year undergraduate course on Functions of Complex Variables and Linear Algebra)**, *BML Munjal University, India*

2015 & 2016 **MATH111 (First year undergraduate course on Calculus and Differential Equations)**, *BML Munjal University, India*

2015 **MAX101 (First year undergraduate course on Differential Equations and Integral Transforms)**, *National Institute of Technology Jalandhar, India*

- 2015 **MAX202 (Second year undergraduate course on Complex Analysis, Series Solution and Calculus of Variation)**, *National Institute of Technology Jalandhar*, India
- 2008–2009 **Topology & Measure Theory (First year masters course)**, *Seth ML(PG) College, Jhunjhunu*, India
- 2008–2009 **Functional Analysis (Second year masters course)**, *Seth ML(PG) College, Jhunjhunu*, India

Teaching Assistant

- 2013 **MA101 (First year undergraduate course on Linear Algebra and Calculus)**, *Indian Institute of Technology Guwahati*, India
- 2012 & 2013 **MA102 (First year undergraduate course on Vector Calculus and Ordinary Differential Equations)**, *Indian Institute of Technology Guwahati*, India
- 2010 & 2012 **MA201 (Second year undergraduate course on Complex Analysis and Partial Differential Equations)**, *Indian Institute of Technology Guwahati*, India

Technical Skills

- Operating sys. Windows, Linux
- Software packages L^AT_EX, Matlab, Maple, GeoGebra, Python

Awards

- 2008 & 2009 **Qualified National Eligibility Test (NET)**, conducted jointly by Council of Scientific & Industrial Research and University Grant Commission, Government of India.
- 2009 **Qualified Graduate Aptitude Test in Engineering (GATE)**, conducted jointly by Indian Institute of Science & seven Indian Institutes of Technology on behalf of Department of Higher Education, Ministry of Human Resources Development, Government of India.
- 2009–2011 **Junior Research Fellowship**, provided by Department of Higher Education, Ministry of Human Resources Development, Government of India.
- 2011–2014 **Senior Research Fellowship**, provided by Department of Higher Education, Ministry of Human Resources Development, Government of India.
- 2017–2018 **NBHM Postdoctoral Fellowship**, provided by National Board for Higher Mathematics, Department of Atomic Energy, Government of India.
- 2018–2021 **Dr. D.S. Kothari Postdoctoral Fellowship**, provided by University Grants Commission, Government of India.

Current Professional Memberships

- Society for Industrial and Applied Mathematics (SIAM)
- Society for Mathematical Biology (SMB)

Other Activities

- Math Club Founder of the Math Club at NIIT University. The club organises problem-solving sessions, quizzes, mathematical games, and application-oriented talks to inspire students and promote exchange of ideas among students and faculty.
- B.Tech. R&D Supervise third-year B.Tech. R&D projects on topics such as Optimum Transport Network Design, Dismantling Corruption/Terrorist Networks, and Game-Theoretic Approaches for Cooperative Electric Vehicle Charging and Routing.
- Guidance
- Applied Innovation Collaborating on the E-Cracker project — an electronic system that replicates traditional firecrackers using sound and light without combustion or pollution. Funding proposal submitted to the Ministry of MSME, Government of India.

References

1. Prof. Soumyendu Raha. *Postdoc Supervisor*
(Reference Letter is available with me)
Department of Computational & Data Sciences, IISc Bengaluru, India
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2. Prof. Ajay Kumar Singh
Area of Electronics and Communication Engineering, NIIT University, Neemrana, Rajasthan
Email: ajay.singh@niituniversity.in, Phone: +91-7408007111.
3. Prof. Ryusuke Kon
Faculty of Engineering, University of Miyazaki, Japan
Email: konr@cc.miyazaki-u.ac.jp, Phone: +81-985-58-7379.
4. Prof. Rajen Kumar Sinha
Department of Mathematics, IIT Guwahati, India
Email: rajen@iitg.ernet.in, Phone: +91-361-2582612.
5. Dr. Sriparna Bandopadhyay (This reference is regarding Teaching Assistant responsibilities at IIT Guwahati)
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